



Optical Sorting of Nanorods via Chiral Dielectric Hyperbolic Metamaterials in Clustered Configurations

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Abstract. In pharmaceutical industries, the classification of plasmonic, dielectric, and chiral nanorods is traditionally achieved through sophisticated laboratory setups, chemical treatments, and multi-step processes. This study introduces an innovative and unique optical sorting technique for nanorods with significant optical force having ~10-20N ranges, enabling the classification of nanorods based on their distinct optical force responses. Pushing effects are generally experienced by objects interacting with light; however, pulling forces or lateral movements can also be exhibited, depending on the interaction. Notably, unique force-direction behaviors are generated for different nanorods simultaneously using a single experimental setup. The proposed optical sorting mechanism is demonstrated using Chiral Dielectric Hyperbolic Metamaterials (CDHMMs) as a substrate. A linearly polarized plane wave, incident at a 90-degree angle, is used to generate distinct responses from three types of nanorods: SiO₂ (dielectric), chiral nanorods, and Au (plasmonic). Lateral forces are displayed by dielectric and chiral nanorods along distinct axes, with the forces on dielectric nanorods being less pronounced. Plasmonic nanorods exhibit substantial optical pushing forces. These effects are caused by the interplay of material characteristics, such as refractive index, chirality, and plasmonic resonance, with the incident light. Through this approach, a sustainable and environmentally friendly alternative to conventional chemical-based sorting techniques is provided, reducing reliance on toxic substances and expensive processes.

Keywords: Optical Sorting, Nanorod, plasmonic, Hyperbolic Metamaterial.

1 Introduction

Optical micromanipulation involves manipulating microscopic objects, such as those in biophysics & cell biology, using light and nanotechnology by transferring momentum toward objects due to radiation pressure along with wave propagation [1]. Moving objects perpendicular to the wave propagation, known as optical lateral forces, were investigated before [2]. Traditional sorting techniques for nanoscale objects are costly chemical approaches [3], specifically in pharmaceuticals. In contrast, using optical

lateral forces on different sizes, shapes, or compositions of objects can be an alternative, cost-efficient way known as optical sorting force. In previous studies, though there have been optical lateral forces on chiral helix particles [4], pulling of plasmonic nanorods [5], optical tweezers approaches [6], plasmonic mode hybridization, and CLOCK microscopy [7] methods remain observed having spacing, size, and orientation sensitivity to nanorods face limitations sorting forces of nanorods remain unexplored, where a brief key comparison with recent works are highlighted in table 1. Without using chemicals, an unstructured light-beam-based approach to nanorod sorting achieves distinct force directions for all nanorods (plasmonic, dielectric, chiral) and could fill the gap in advanced material science and biology. In this article, multiple nanorods display pushing and lateral force behavior across three axes (x, y, z) as optical sorting.

The main objective of this study is to develop a novel technique to sort out the three distinct nanorods, dielectric, chiral, and plasmonic, based on their distinct optical force response. Due to the reversal current density of plasmonic nanorods, they experience pushing forces. While chiral and dielectric lateral forces are due to the dipole electric field density and the Poynting vector. Further research may be needed for arbitrary sorting for effective cluster combinations. This environment-friendly technique can be enhanced, efficient, and precise in the pharmaceutical industry to sort gold nano-rods, microplastics, and chiral molecules.

2 Optical Setup and Methods

Initially, the proposed optical setup was arranged in a water medium, comprising a cluster of three different nanorods: dielectric (SiO₂), plasmonic (Au), and chiral, placed over a chiral dielectric hyperbolic metamaterial (CDHMM) substrate, as shown in Fig. 1, to achieve optical sorting. Secondly, the incident magnetic field of the plane-polarized, non-structured, time-harmonic light beam propagates from “+Z” to “-Z” direction with polarization in the “Y” plane can be expressed as:

$$H = H_0 e^{ikz} \hat{y} \quad (1)$$

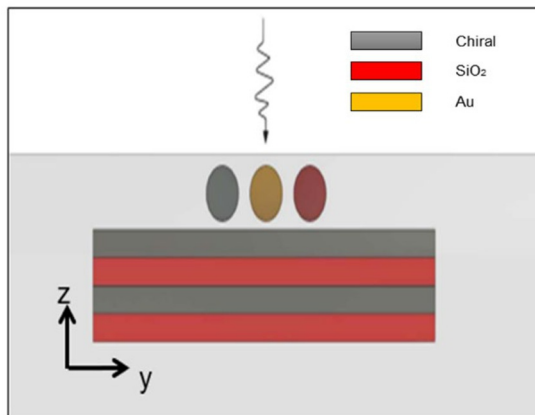


Fig. 1. Illuminating a schematic view of our proposed optical setup immersed in a water medium as a 2D view, where a plane-polarized optical beam propagates along the “+z” to “-z” direction.

Note that water is thought to have a refractive index of “ $n = 1.33$.” The proposed periodic bilayer type, period “ $p = 40$ nm”, hyperbolic metamaterial of chiral dielectric having a 600nm length & width of each layer with thickness “ $t = 20$ nm”. 150 nm and 30 nm are the length and radius of each identical nanorod. The gap between two adjacent nanorods is set to 25 nm. The distance from the upper surface of the suggested hyperbolic metamaterial to the cluster's lower surface is $h = 30$ nm. The actual part of the refractive index of the nanorod (NR) of dielectric and layer in the CDHMM of dielectric (SiO₂) has been defined in the simulation set as “ $n = 1.45$ ”. Lastly, the full-wave simulation, including optical force, is observed from 780 nm to 1600 nm with a 5 nm step size of incident wavelength “ λ ” and was computed using the COMSOL Multiphysics 6.0 numerical solver.

3 Theory

The forces calculated outside and inside the nanorod volume in the full wave simulations are known as exterior and interior forces, respectively. The time-averaged Minkowski stress tensor is used to derive the exterior force at $r = a + (3$ nm away from the surface outside), where “ a ” is the radius of the background field's scatterer:

$$\langle F_{\text{Total}} \rangle = \oint [\mathbf{T}^{\text{out}}] \cdot d\mathbf{s} \quad (2)$$

Where $\mathbf{T}^{\text{out}}$ can be expressed as,

$$\langle \mathbf{T}^{\text{out}} \rangle = \frac{1}{2} [(D_{\text{out}} E_{\text{out}}^* + B_{\text{out}} H_{\text{out}}^*) - \frac{1}{2} I(E_{\text{out}}^* \cdot D_{\text{out}} + H_{\text{out}}^* \cdot B_{\text{out}})] \quad (3)$$

4 Results and Discussions

Here, depicted in Fig. 2, dielectric nanorods exhibit lateral force (-y direction) from 780 nm to 1390 nm, and plasmonic nanorods show a pushing force (-z-direction) from 780 nm to 1400nm, and chiral nanorods experience lateral force (+x direction) from 780 nm to 1130nm.

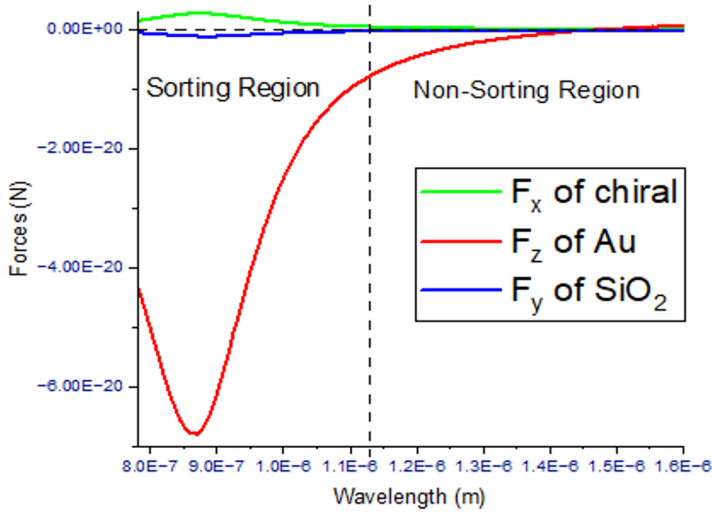


Fig. 2. shows their corresponding optical force.

4.1 Sorting of chiral & dielectric nanorod: the origin of the optical lateral force

The electric field distribution has been modeled on chiral and dielectric nanorods due to the interference of waves in between them, used to show the lateral force along the distinct directions. The induced electric dipole (ED) and magnetic dipole (MD) moments for the objects are known, and the force (time-averaged) experienced by each NR can be defined as,

$$\langle F \rangle = \frac{1}{2} \text{Re} \left[p(\nabla \otimes E^*) + m(\nabla \otimes H^*) - \left(\frac{ck^4}{6\pi} \right) \left(\frac{1}{\sqrt{\epsilon_\beta \mu_\beta}} \right) (p \times m^*) \right] \quad (4)$$

Here, p and m are the electric and magnetic dipole moments, respectively. The optical lateral force in chiral and dielectric nanorods mainly originates from the induced electric dipole, magnetic dipole moment, and interaction between the electric and magnetic dipole moments.

$$\langle F \rangle = \langle F_e \rangle + \langle F_m \rangle + \langle F_{e-m} \rangle \quad (5)$$

Where $\langle F_e \rangle$, $\langle F_m \rangle$, and $\langle F_{e-m} \rangle$ represent the force induced due to electric dipole, magnetic dipole, and interaction between electric and magnetic dipoles, respectively. Here, the $\langle F_e \rangle$ portion is mainly responsible for the lateral force in the chiral and dielectric nanorod. The induced electric dipole distributions, the chiral and dielectric nanorods, show the optical lateral forces in the sorting region (780~1130nm) shown in Fig. 3(a & c). From ref [1], Spin-orbit coupling and chirality-dependent interaction drive the asymmetric (anticlockwise) behavior, shown in Fig. 3(g), exhibiting the chiral nanorod's

lateral forces along the x-axis, while dielectric shows (clockwise) asymmetric behavior plays a role in directing forces along the y-axis, shown in Fig. 3(h).

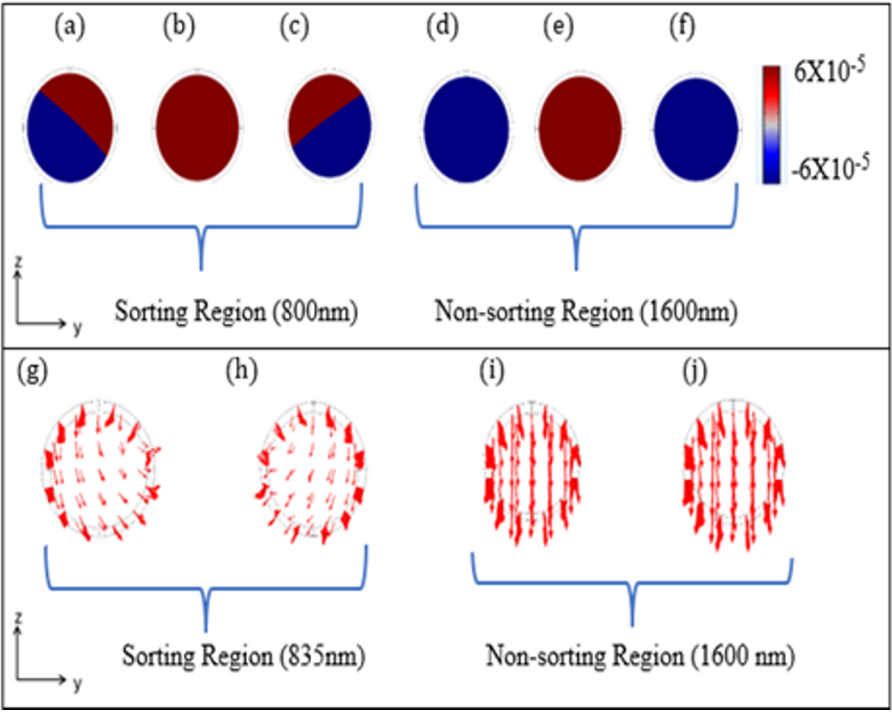


Fig. 3. Illustrates electric field distribution in (a) and (c), where an electric dipole is induced on the chiral and dielectric nanorod at the sorting region, respectively. (d) and (f) reveal that the electric field has no impact on the chiral and dielectric nanorod (non-sorting region), respectively. On the other hand, (b & e) display no changes on the plasmonic nanorod at sorting (800 nm) and non-sorting region (1600 nm). Poynting vector arrows are demonstrated in (g) and (h) at the sorting region (< 1130 nm) of chiral and dielectric nanorod with an asymmetric manner with a reversal direction to each other, while (i) and (j) display each of which arrows in the -Z axis direction as the non-sorting region.

These different directions of asymmetric Poynting vectors between them, chiral & dielectric nanorods, lead to their distinct direction of lateral forces. On the other hand, the absence of induced electric dipoles and symmetric Poynting vector flow along the -z-axis of chiral and dielectric nanorods results in negligible lateral forces in the non-sorting region (1130~1600 nm) that is shown in Fig. 3(d & f) and Fig. 3(i & j), respectively. For this same reason, the plasmonic found no lateral forces shown in Fig. 3(b & e). Thus, we can say that both chiral and dielectric nanorods' lateral forces with respective distinct directions mainly originated from the induced electric dipole and asymmetric Poynting vector contribution.

Table 1. Comparison with recent research.

Article	Materials	Setup	Results
[2]	Chiral(c), plasmonic(p), dielectric(d) particles	Placed on a plasmonic substrate	Optical Sorting Forces, nearly 10^{-26} N
[3]	Plasmonic nanowire	Placed on p-d-p-d substrate	Optical Pulling Forces at Pico Scale
[1]	Helix, spherical particles	Placed on a metallic and dielectric substrate	Optical Lateral Forces
This work	Chiral, dielectric, and plasmonic nanorods	Placed on c-d-c-d substrate	A significant Optical Sorting at 10^{-20} N

4.2 Sorting of plasmonic nanorod: the origin of the optical pushing force

According to the ref [11], the distribution of induced current densities within active plasmonic scatterers is directly linked to both bound charges and currents of the force contribution. When force reversal takes place, the current density of gold nanorods occurs around the region, which is shown in Figs. 4(a) and 4(b). The force densities on bound charges and current, either positive or negative, result in pushing and pulling forces. The following equation may also describe the pushing and pulling force on the plasmonic nanorod:

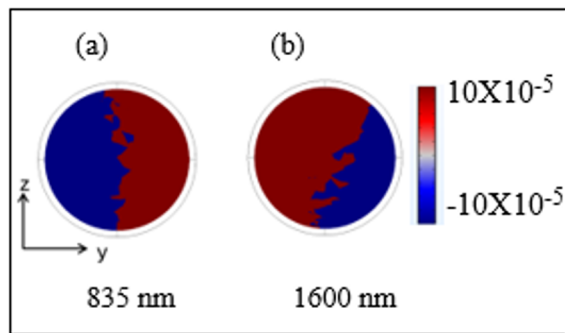


Fig. 4. Illustrates the Current densities (J_z) distribution of plasmonic nanorod placed over our proposed configuration at an incident beam representing wavelengths of (a) 835 nm (sorting region), and (b) 1600 nm (non-sorting region).

$$\langle F_{\text{total}} \rangle = \frac{1}{2} \text{Re} \left[\oint \left(-\frac{1}{2} |E_{\text{Surf}}|^2 \nabla \epsilon \right) ds \right] + \frac{1}{2} \text{Re} \left[\int (\rho_{\text{free}} E_{\text{in}} + J_{\text{free}}(\text{in}) \times \mu_0 H_{\text{in}}) dv \right] \quad (6)$$

Here, “surf” and “in” denote the scatterer's surface and the scatterer's interior, respectively. Using Equation (2), we can rewrite Equation (6) as

$$\oint \langle \mathbf{T}^{\text{out}} \rangle \cdot d\mathbf{s} = \mathbf{F}_{\text{total}} = \langle \mathbf{F}_{\text{Mink}}^{\text{Surface}} \rangle + \langle \mathbf{F}_{\text{Mink}}^{\text{Bulk}} \rangle \quad (7)$$

For lossy objects like plasmonic nanorods, the bulk force, strongly influenced by free current density, in equation (7) is never zero, resulting in a time-average pushing force. When force reversal occurs, current density changes, which are aligned with the equation (7), prove the optical pushing forces throughout the sorting region shown in Fig. 4(a & b).

5 Conclusion

This study shows that optical sorting may be applied to a heterogeneous cluster of dielectrics, plasmonic, and chiral nanorods in a water medium using an optical setup that uses a chiral dielectric hyperbolic metamaterial. We found that the three-nanorod cluster was subjected to a high sorting force at incoming light wavelengths between 780 and 1130 nm. As far as we are aware, there is currently no literature that demonstrates the capacity to use this process to draw any kind of object, including clustered nanorods, such as chiral, plasmonic, and dielectric nanorods, using a single optical setup. This thorough analysis can offer possible uses and suggestions for further research in this field.

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Disclosure of Interests. The authors declare that there are no conflicts of interest related to this work.

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